SLEEP HEALTH AND LÏFESTYLE ANALYSIS

Name = Simarpreet Kaur

Roll no = 2420432

Course = B.Tech CSE(DS)

PROJECT OVERVIEW

Objective: To analyze the relationship between daily lifestyle choices and sleep health metrics using a multi-dimensional dataset.

Importance: Sleep is a fundamental pillar of health. Understanding the drivers of poor sleep can help in developing personalized wellness interventions.

Dataset Source: A comprehensive survey of 374 individuals covering 13 health and behavioral features.

DATASET DESCRIPTION

Sleep Duration	Total hours of sleep per night	5.8 - 8.5 hours
Quality of Sleep	Subjective rating of sleep restfulness	4 - 9 (Scale)
Stress Level	Self-reported daily stress intensity	3 - 8 (Scale)
Physical Activity	Minutes of activity per day	30 - 90 minutes
Daily Steps	Total step count per day	3,000 - 10,000

DATA PREPARATION

Cleaning

Checked for missing values and handled categorical encoding for BMI and Occupation.

Normalization

Applied Min–Max scaling to Duration, Quality, Stress, and Activity to create a 0–1 range.

Sleep Score

Engineered a composite "Sleep Score" as a weighted average of normalized lifestyle factors.

OVERALL DATASET INSIGHTS

374

Total Participants Analyzed

Mean Sleep Quality: 7.31/1O

Average Heart Rate: 70.1 bpm

7.1h

Average Sleep Duration

Mean Stress Level: 5.38/10

Average Daily Steps: 6,816 steps

BEST SLEEPERS ANALYSIS

Top 5 Participant Scores

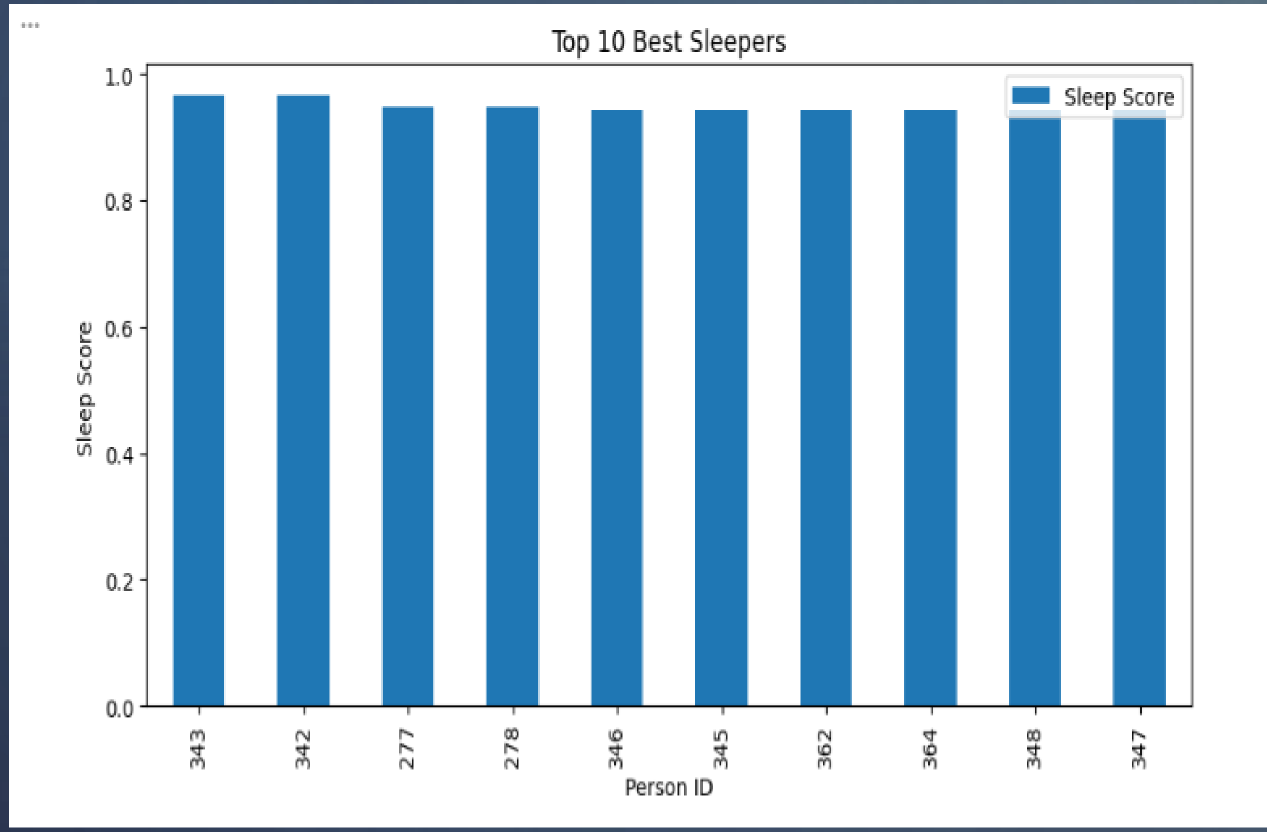
ID 342 (Doctor)

ID 343 (Doctor)

ID300 (Engineer)

ID 301 (Engineer)

ID302 (Engineer)



Key Traits of Best Sleepers

- High Physical Activity (Avg: 80 min/day)
- Low Stress Levels (Avg: 3.0/10)
- Healthy Heart Rate (Avg: 71 bpm)
- Balanced BMI Category

	0
Sleep Duration	8.18
Quality of Sleep	9.00
Stress Level	3.00
Physical Activity Level	80.00
Daily Steps	6940.00
Heart Rate	71.00

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Worst Sleepers Analysis

Bottom 5 Participant Scores

ID 81 (Scientist)

ID 82(Scientist)

ID 4 (Sales Rep)

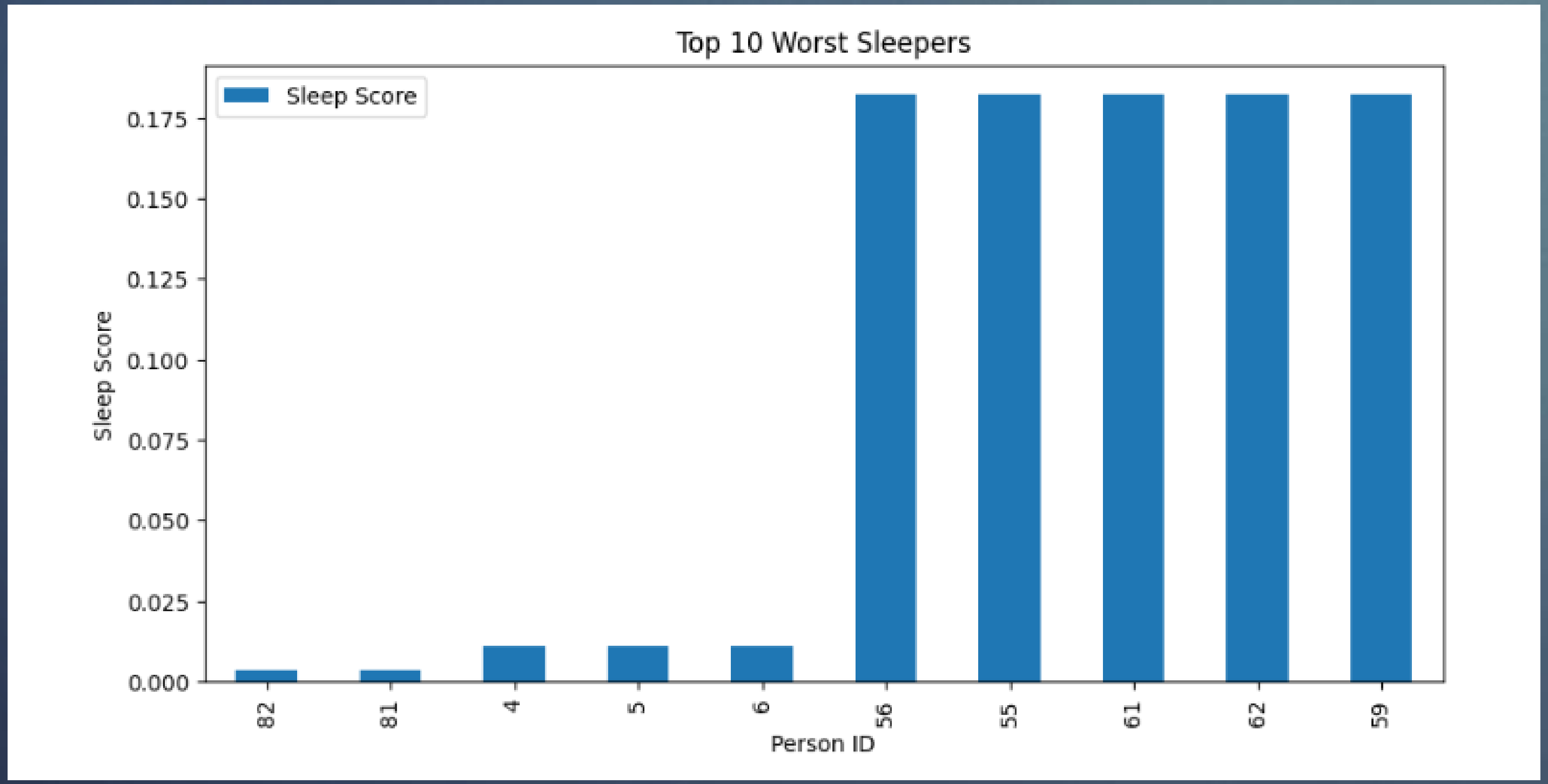
ID 5 (Sales Rep)

ID 6 (SW Engineer)

Primary Risk Factors

- Short Sleep Duration (Avg: 5.9h)
- Peak Stress Levels (Avg: 8.0/10)
- Low Physical Activity (Avg: 30 min)
- BMI Category: Often Overweight/Obese

The worst sleeper (ID 81) had a score of nearly zero, impacted by Sleep Apnea.



	0
Sleep Duration	5.93
Quality of Sleep	5.00
Stress Level	8.00
Physical Activity Level	30.40
Daily Steps	4440.00
Heart Rate	77.70

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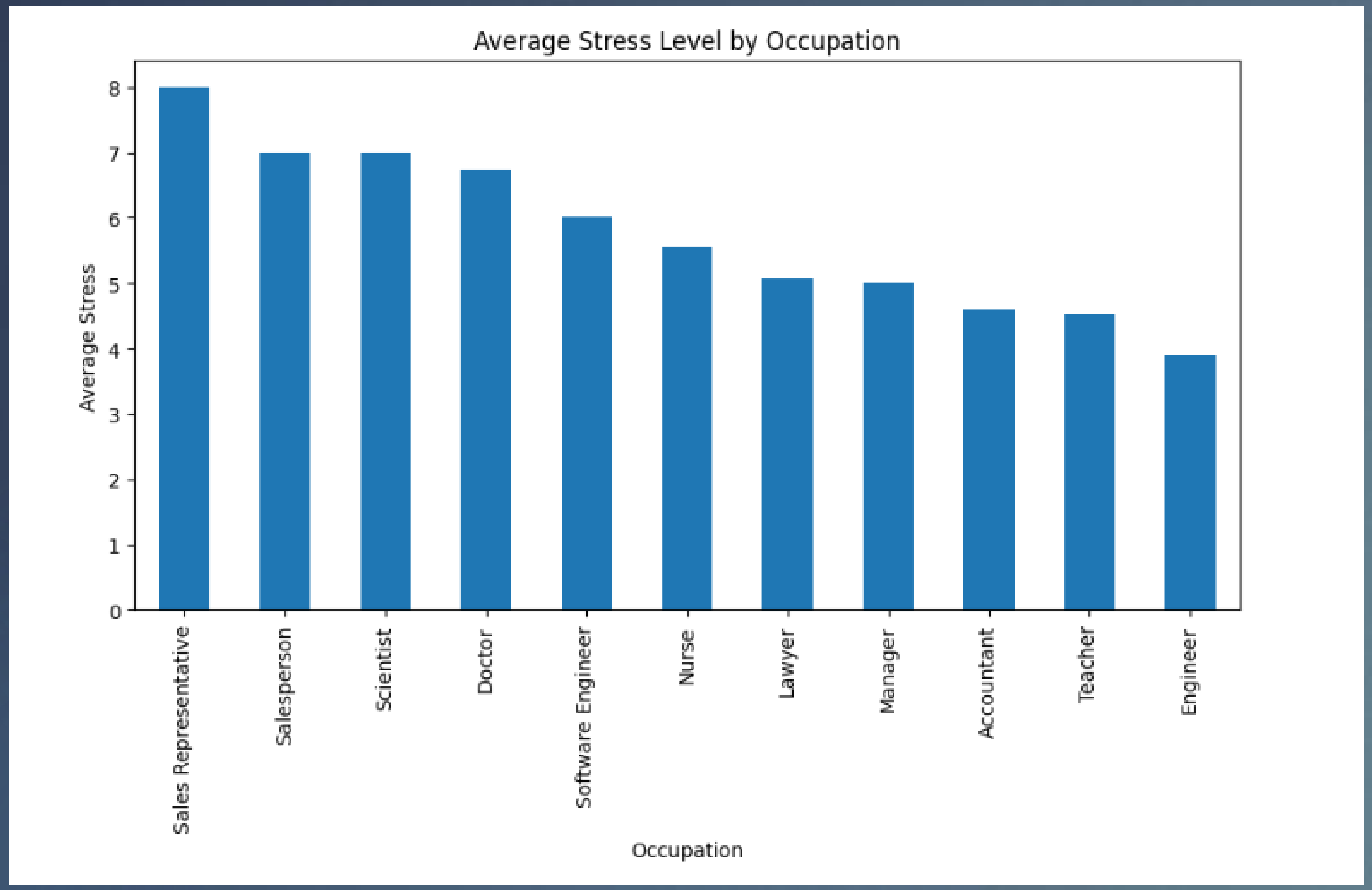
STRESS LEVEL ANALYSIS

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...          Sleep Duration  Quality of Sleep  Stress Level  \
  Gender
  Female          7.229730          7.664865      4.675676
  Male            7.036508          6.968254      6.079365

          Physical Activity Level
  Gender
  Female                59.140541
  Male                  59.201058
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Impact on Sleep Quality

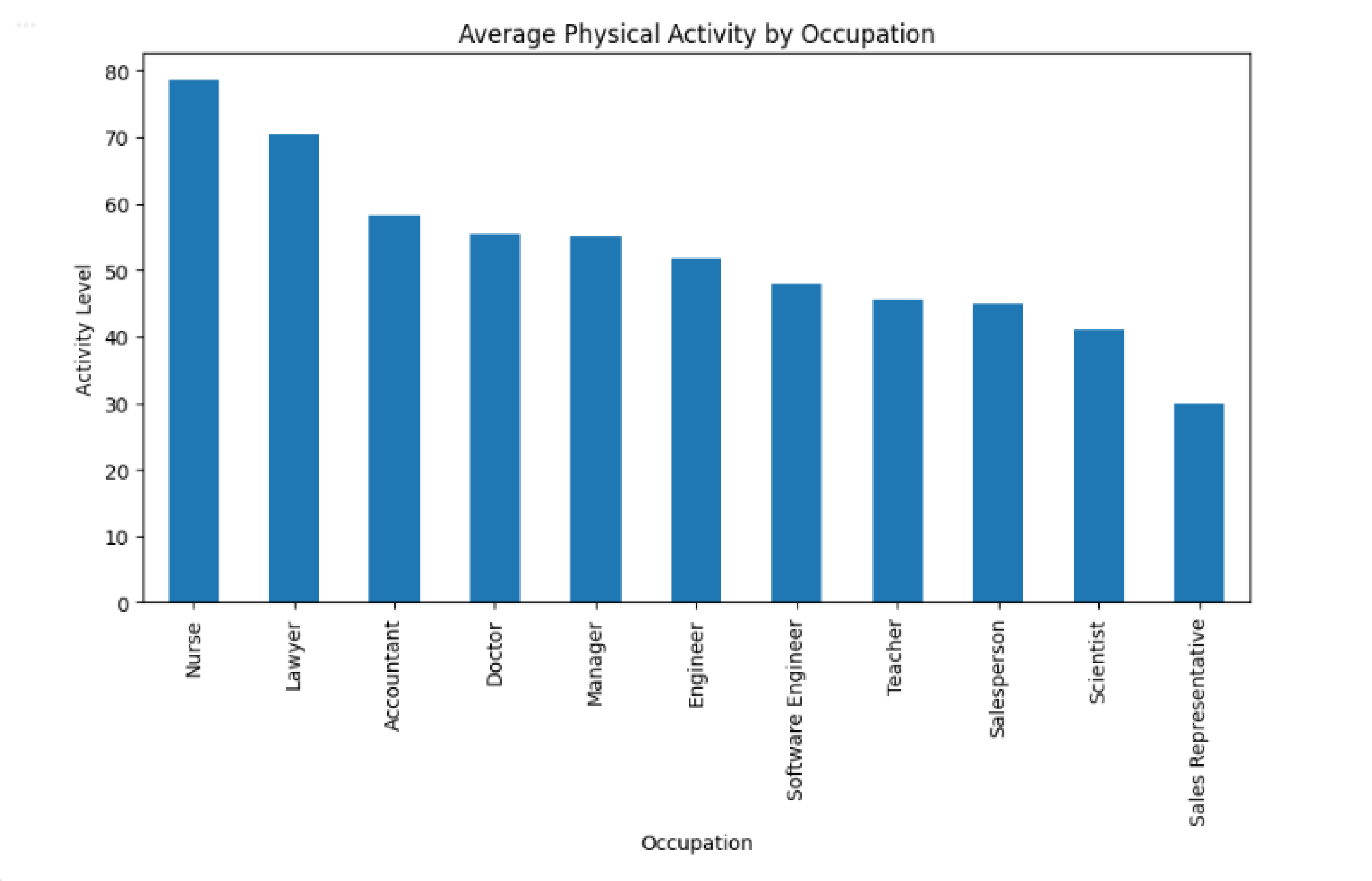
The analysis reveals a strong negative correlation (-0.90) between Stress Level and Quality of Sleep.

3 (Low)	8.5
5 (Medium)	7.0
8 (High)	4.5

PHYSICAL ACTIVITY ANALYSIS

Activity & Sleep Synergy

While stress degrades sleep, activity builds it. Individuals with ›7O min of activity showed significantly higher restfulness.

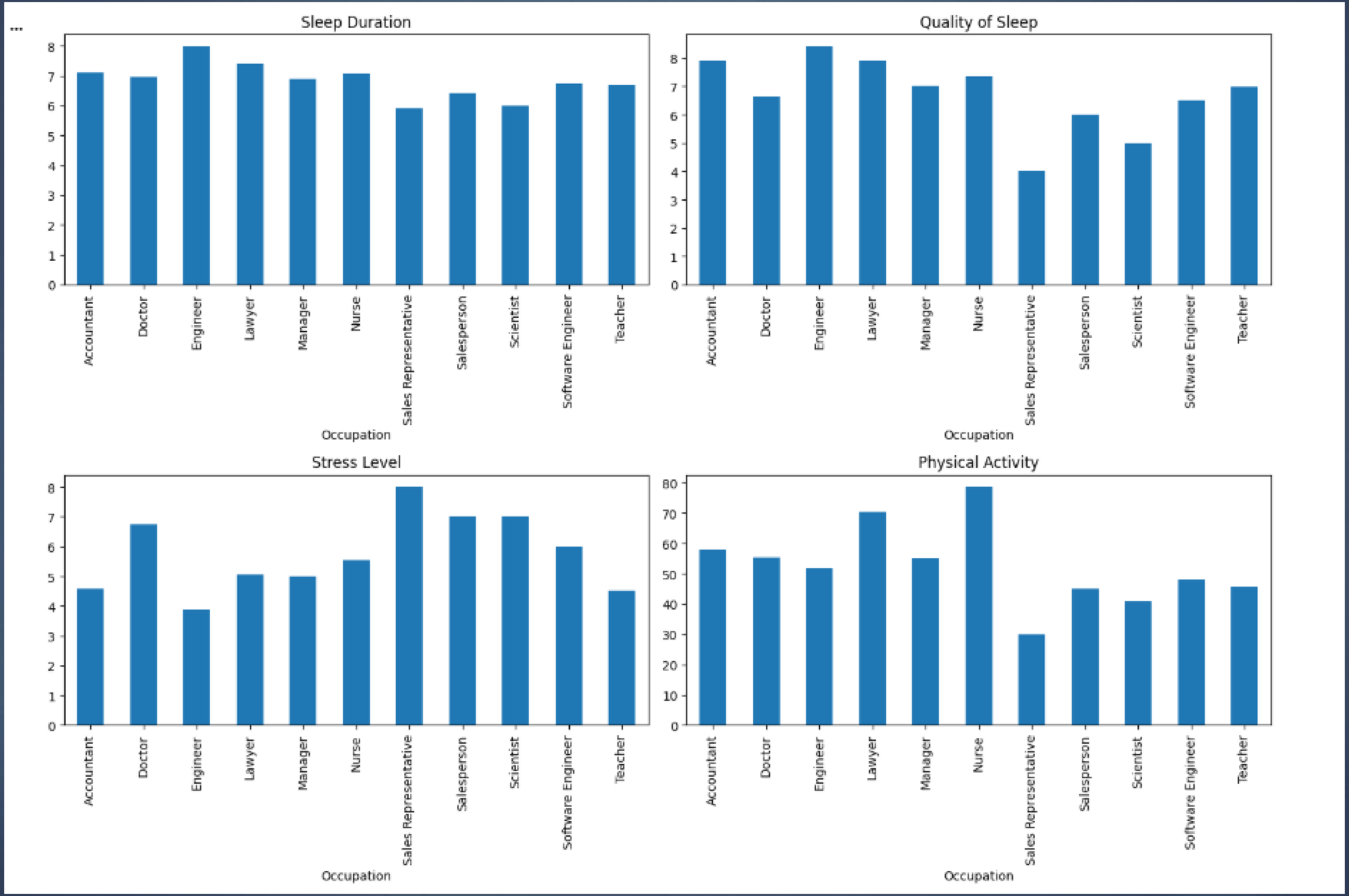


't Daily Steps: Higher step counts (8k—10k) correlate with lower resting heart rates (65—70 bpm).

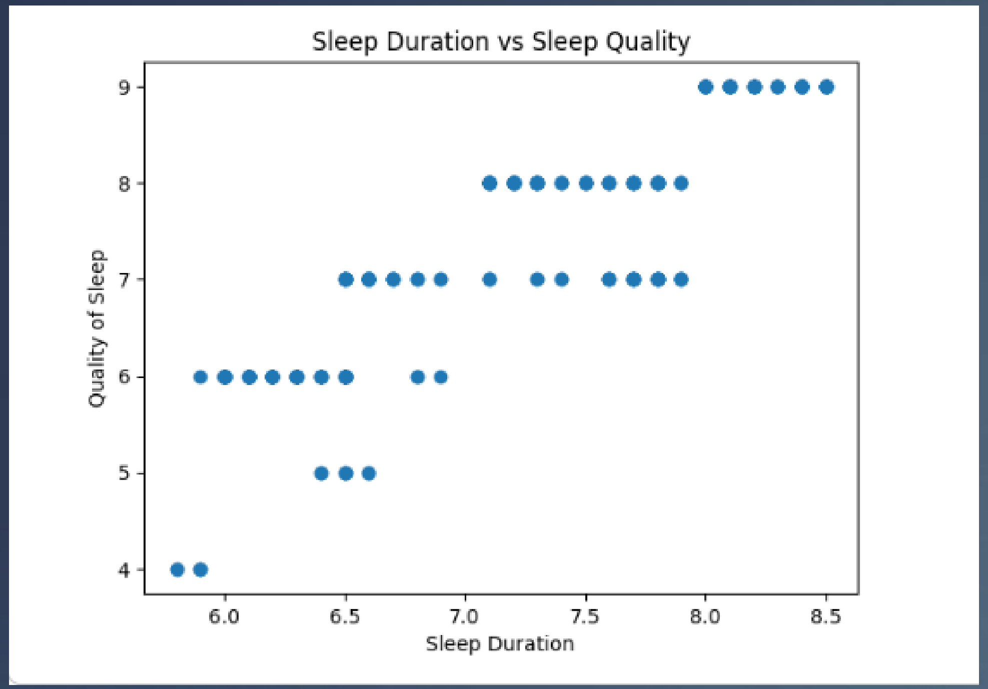
" Cardio Health: Consistent activity keeps the heart rate stable, reducing nighttime awakenings.

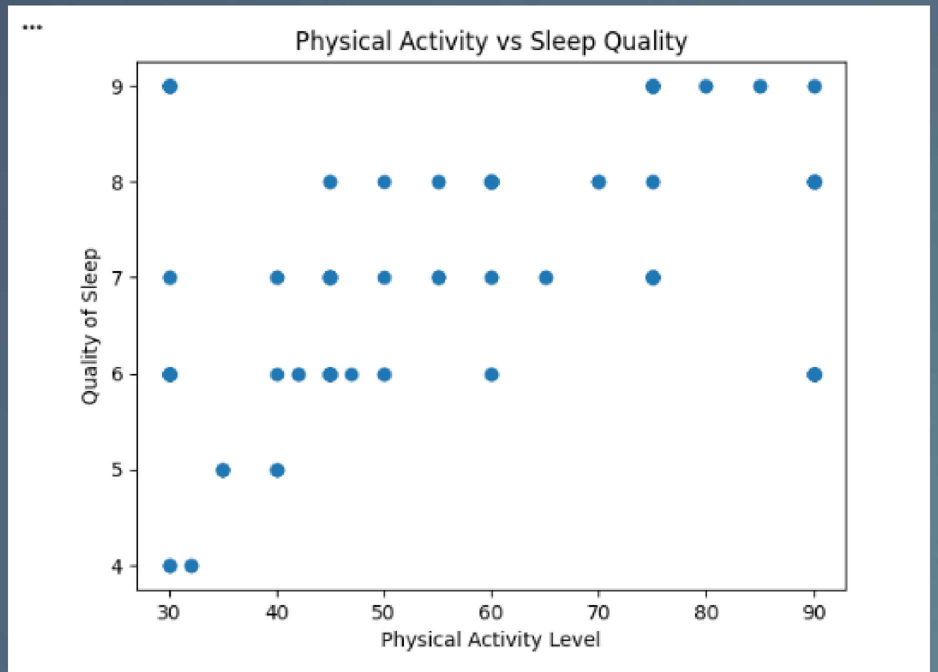
Occupation	Physical Activity Level
Accountant	58.108108
Doctor	55.352113
Engineer	51.857143
Lawyer	70.425532
Manager	55.000000
Nurse	78.589041
Sales Representative	30.000000
Salesperson	45.000000
Scientist	41.000000
Software Engineer	48.000000
Teacher	45.625000

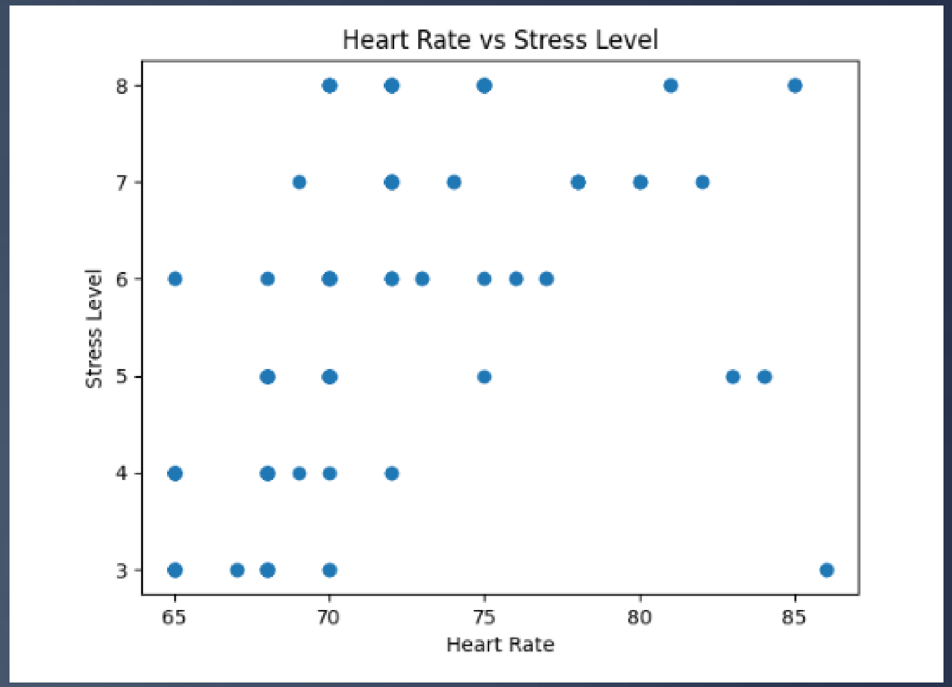
Occupation Analysis



Correlation Analysis







Quality of Sleep vs. Sleep Duration	+0.88	Strong Positive
Heart Rate vs. Stress Level	+0.60	Moderate Positive
Quality of Sleep vs. Daily Steps	-0.65	Moderate Negative
Physical Activity vs. Quality of Sleep	+0.75	Strong Positive

BEST LIFESTYLE

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder	Sleep_Duration_Norm	Quality_Norm	Stress_Norm	Activity_Norm	Sleep Score
32	33	Female	31	Nurse	7.9	8	75	4	Normal Weight	117/76	69	6800	NaN	0.777778	0.8	0.8	0.75	0.788333
50	51	Male	32	Engineer	7.5	8	45	3	Normal	120/80	70	8000	NaN	0.629630	0.8	1.0	0.25	0.733889
51	52	Male	32	Engineer	7.5	8	45	3	Normal	120/80	70	8000	NaN	0.629630	0.8	1.0	0.25	0.733889
85	86	Female	35	Accountant	7.2	8	60	4	Normal	115/75	68	7000	NaN	0.518519	0.8	0.8	0.50	0.685556
86	87	Male	35	Engineer	7.2	8	60	4	Normal	125/80	65	5000	NaN	0.518519	0.8	0.8	0.50	0.685556

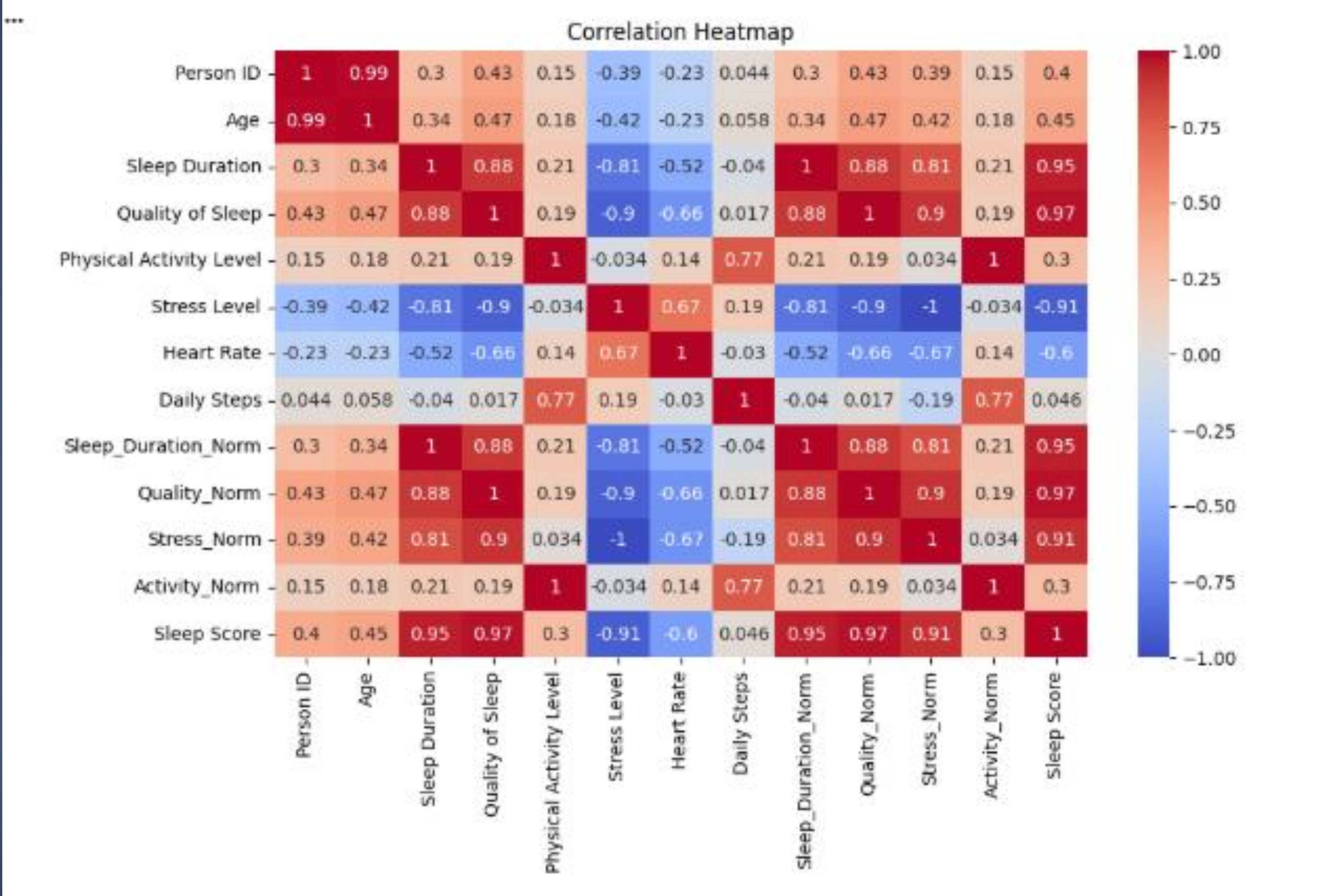
WORST LIFESTYLE

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder	Sleep_Duration_Norm	Quality_Norm	Stress_Norm	Activity_Norm	Sleep Score
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea	0.037037	0.0	0.0	0.000000	0.011111
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea	0.037037	0.0	0.0	0.000000	0.011111
5	6	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia	0.037037	0.0	0.0	0.000000	0.011111
16	17	Female	29	Nurse	6.5	5	40	7	Normal Weight	132/87	80	4000	Sleep Apnea	0.259259	0.2	0.2	0.166667	0.214444
18	19	Female	29	Nurse	6.5	5	40	7	Normal Weight	132/87	80	4000	Insomnia	0.259259	0.2	0.2	0.166667	0.214444

KEY FINDINGS

- Stress is the Primary Killer: It has the highest negative impact on sleep restfulness across all groups.
  - Duration vs. Quality: While they are linked, high duration does not always guarantee high quality if stress is present.
  - The “Engineer" Advantage: Certain professions provide structural stability that promotes better sleep hygiene. Physical Activity Buffer: 60+ minutes of activity helps mitigate some effects of moderate stress.

OVERALL CORRELATION HEATMAP



CONCLUSION

Sleep health is not an isolated metric; it is the outcome of a complex interplay between professional demands, physical habits, and psychological state.

By optimizing activity and aggressively managing stress, individuals can significantly improve their Quality of Sleep, regardless of their age or gender.

Thank You!